

Real Python Claude Code Cheat Sheet

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Install and Start

- Start `claude` in your project root and avoid your home folder.
- Sign in with a paid Claude plan or an Anthropic Console account.
- Type `/` for commands and `?` for shortcuts.

Install on macOS and Linux

```
$ curl -fsSL https://claude.ai/install.sh \
    | bash
```

Install on Windows

```
PS> irm https://claude.ai/install.ps1 | iex
```

Check and Update

```
$ claude --version
$ claude update
```

Start a Session

```
$ cd my-project/
$ claude                      # Interactive
$ claude "explain this repo"  # With prompt
$ claude -n auth-fix          # Named
```

Resume a Session

```
$ claude -c          # Latest one in this dir
$ claude -r          # Pick from a list
$ claude -r auth-fix "finish the PR"
```

Launch Flags

```
$ claude --model sonnet
$ claude --effort high
$ claude --permission-mode plan
$ claude --add-dir ../shared-lib
$ claude -w feature-x   # In a git worktree
```

Walk through a full project in How to Use Claude Code to Write and Debug Python.

Learn More on realpython.com/search:
`claude code` · `agentic coding`

Run Claude Code Headless

- `-p` prints one answer and exits, so it fits scripts and CI.
- `--allowedTools` pre-approves the tools a script needs.
- `--bare` skips CLAUDE.md, hooks, skills, and MCP for faster starts.

One-Shot Prompts

```
$ claude -p "explain this function"
$ cat error.log | claude -p "root cause?"
$ claude -c -p "check for type errors"
```

JSON Output and Limits

```
$ claude -p "summarize this project" \
    --output-format json | jq -r '.result'
$ claude -p "fix the failing test" \
    --allowedTools "Read,Edit,Bash" \
    --max-turns 5
```

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`claude code` · `continuous integration`

Session Slash Commands

- Type `/` to list commands, then keep typing to filter.
- Put the command at the start of your message.
- Run `/clear` when you switch tasks and `/compact` to keep going.

Context

Command	What it does
`/context`	Show context usage
`/compact [focus]`	Summarize the chat to free space
`/clear`	Start a fresh chat. Also `/new`
`/btw [question]`	Ask a side question

Sessions

Command	What it does
`/resume [name]`	Reopen a session. Also `/continue`
`/rename [name]`	Label this session
`/branch [name]`	Try another direction
`/rewind`	Roll back code and chat. Also `/undo`
`/export [file]`	Save the chat as plain text
`/exit`	Quit. Also `/quit`

Background Work

Command	What it does
`/tasks`	List running shells and agents
`/subtask <task>`	Hand a task to a forked subagent
`/fork [prompt]`	Run a copy in the background
`/background`	Detach the session. Also `/bg`
`/goal [condition]`	Keep working until the goal is met
`/loop 5m <prompt>`	Repeat a prompt on an interval

Learn More on realpython.com/search:
`context window` · `claude code`

Setup and Review Slash Commands

- Press `s` in the `/model` picker to switch for this session only.
- Raise `/effort` for hard problems, at the cost of more tokens.

Model and Usage

Command	What it does
`/model [name]`	Switch model and save it as default
`/effort high`	Set effort from low to max, or auto
`/fast`	Toggle fast mode, or pass `on` or `off`
`/usage`	Show cost and limits. Also `/cost`

Configure

Command	What it does
`/config`	Open settings. Also `/settings`
`/config theme=dark`	Set one key directly
`/permissions`	Edit allow, ask, and deny rules
`/memory`	Edit memory files
`/hooks`	View configured hooks
`/skills`	List and toggle skills
`/mcp`	Manage MCP servers and OAuth
`/doctor`	Diagnose your setup
`/status`	Show version and account

Plan and Review

Command	What it does
`/init`	Draft a CLAUDE.md
`/plan [task]`	Enter plan mode
`/diff`	Review the working tree
`/code-review --fix`	Find and fix bugs
`/code-review ultra`	Run a deep cloud review
`/security-review`	Find security issues
`/simplify`	Clean up the code

Learn More on realpython.com/search:
`code review` · `claude code`

More Slash Commands

- Your skills and plugin skills share the same `/` menu.
- MCP server prompts show up as `/server:prompt`.

Workspace

Command	What it does
`/add-dir <path>`	Grant access to another directory
`/cd <path>`	Move the session and keep the chat
`/ide`	Connect your IDE
`/copy [N]`	Copy a response
`/insights`	Report on your sessions

Look and Feel

Command	What it does
`/theme`	Pick a color theme
`/color [color]`	Set the prompt bar color
`/statusline`	Configure the status line
`/keybindings`	Edit the shortcuts file
`/help`	List all commands
`/release-notes`	See what's new

Extend

Command	What it does
`/plugin`	Browse and install plugins
`/reload-skills`	Pick up new skills
`/workflows`	Watch dynamic workflows
`/schedule`	Set up recurring cloud routines

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`claude code` · `ai coding tools`

Keyboard Shortcuts

- On macOS, press `Option` for `Alt`. Most of these need Option set as Meta, but `Option+T` works without it.

Control the Session

Keys	Action
`Shift+Tab`	Cycle permission modes
`Esc`	Interrupt Claude
`Esc Esc`	Clear input or rewind
`Ctrl+C`	Interrupt, clear, or exit when pressed twice
`Ctrl+O`	Toggle transcript viewer
`Ctrl+B`	Background a running task
`Ctrl+T`	Toggle the task checklist
`Alt+P`	Switch model
`Alt+T`	Toggle extended thinking
`Alt+O`	Toggle fast mode
`Ctrl+L`	Redraw the screen

Write Your Prompt

Keys	Action
`\ Enter`, `Ctrl+J`	New line
`Shift+Enter`	New line in most terminals
`Ctrl+R`	Search prompt history
`Ctrl+G`	Edit prompt in your editor
`Ctrl+S`	Stash or restore the prompt
`Ctrl+V`	Paste an image
`Cmd+V`	Paste an image in iTerm2
`Alt+V`	Paste an image on Windows or WSL
`!` at start	Shell mode
`@`	Mention a file or subagent
`?`	Show all shortcuts

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Real Python

CLAUDE.md and Memory

- CLAUDE.md holds instructions you write. Auto memory holds notes Claude writes.
- All files load together from broad to specific, so remove contradictions.
- Since v2.1.277, `AGENTS.md` loads when a repo has no CLAUDE.md.

Where CLAUDE.md Files Live

File	Scope
`~/.claude/CLAUDE.md`	You, in every project
`./CLAUDE.md`	Your team. Commit it
`./CLAUDE.local.md`	You, in this project
`sub/CLAUDE.md`	Loads on demand

Import Files

```
# Project Notes
See @README.md for an overview.
- Testing: @docs/testing.md
- Mine: @~/.claude/my-notes.md
```

Scope a Rule in .claude/rules/api.md

```
---
paths:
  - "src/api/**/*.py"
---
- Validate every request body
```

Set up global, project, and local files in How to Write a CLAUDE.md File for Claude Code.

Learn More on realpython.com/search:

`claude.md` · `context engineering`

Permissions and Modes

- `Shift+Tab` cycles `default`, `acceptEdits`, and `plan`, plus `auto` when it's available.
- Since v2.1.283, most terminal and VS Code sessions start in `auto`.
- The UI labels `default` mode as Manual.
- Rules check deny, then ask, then allow. The first match wins.
- Deny rules hold in every mode, even `bypassPermissions`.

Mode	Runs without asking
`default`	Reads only
`acceptEdits`	Reads, edits, file commands
`plan`	Reads to draft a plan
`auto`	Everything, with safety checks
`dontAsk`	Pre-approved tools and file reads
`bypassPermissions`	Everything, in isolated VMs

Permission Rules

```
{
  "permissions": {
    "allow": ["Bash(uv run pytest *)"],
    "ask": ["Bash(git push *)"],
    "deny": ["Read(./.env)"]
  }
}
```

Learn More on realpython.com/search:

`claude code` · `ai coding agent`

Settings Files

- Managed settings win, followed by `--settings`, local, project, and user.
- List keys like `permissions.allow` merge across files.
- Settings files are strict JSON with no comments or trailing commas.
- `/status` shows which files loaded.

Where Settings Live

File	Scope
`~/.claude/settings.json`	User
`.claude/settings.json`	Project, shared
`.claude/settings.local.json`	Project, just you

Learn More on realpython.com/search:

`claude code` · `json`

Hooks

- Hooks run your commands at fixed points in Claude's loop.
- Exit code 2 from `PreToolUse` blocks the tool call.
- `/hooks` shows what's registered.

Key Events

Event	When it runs
`SessionStart`	Session begins or resumes
`UserPromptSubmit`	Before Claude sees your prompt
`PreToolUse`	Before a tool runs. Can block it
`PostToolUse`	After a tool succeeds
`Notification`	Claude needs your input
`Stop`	Claude finished replying

Format After Every Edit

```
{
  "hooks": {
    "PostToolUse": [{
      "matcher": "Edit|Write",
      "hooks": [{
        "type": "command",
        "command": "ruff format ."
      }]
    }]
  }
}
```

Go deeper with these hooks resources:

- How to Automate Your Workflow With Claude Code Hooks
- Claude Code Hooks Cheat Sheet

Learn More on realpython.com/search:

`claude code hooks` · `ruff`

Skills

- A folder with a `SKILL.md` becomes a `/name` command.
- Claude also loads a skill when its description fits the task.
- `disable-model-invocation: true` makes a skill manual-only.
- Old `.claude/commands/*.md` files still work.

Where Skills Live

File	Scope
`~/.claude/skills/<name>/SKILL.md`	You
`.claude/skills/<name>/SKILL.md`	Project

A Minimal SKILL.md

```
---
name: docstring-doctor
description: >-
  Writes Google-style Python docstrings.
  Use when asked to document code.
---
Open with a one-line summary...
```

Claude follows the instructions below the frontmatter each time the skill runs.

Learn More on realpython.com/search:

`claude code skills` · `docstrings`

Subagents

- A subagent works in its own context and reports back.
- Built-in subagents include Explore, which searches without editing, and Plan.
- Claude can write one for you, or you can add a file to `.claude/agents/`.

Define .claude/agents/code-reviewer.md

```
---
name: code-reviewer
description: Reviews code for quality
tools: Read, Glob, Grep
model: sonnet
---
You are a code reviewer...
```

Run the Session as an Agent

```
$ claude --agent code-reviewer
```

Call a Subagent

```
@agent-code-reviewer check my changes
```

Learn More on realpython.com/search:

`claude code` · `ai agents`

MCP Servers

- MCP connects Claude Code to outside tools and data.
- Scopes are `local`, `project`, and `user`, with `local` as the default.
- Project scope saves servers to a shared `.mcp.json` file.
- For stdio servers, `--` comes before the server command.
- Connect databases with a read-only user.

Add a Server

```
$ claude mcp add --transport http \
    notion https://mcp.notion.com/mcp
$ claude mcp add --transport stdio \
    --scope project db \
    -- npx -y @bytebase/dbhub \
    --dsn "postgresql://readonly:pw@host/db"
```

Manage Servers

```
$ claude mcp list
$ claude mcp get notion
$ claude mcp remove notion
```

Learn More on realpython.com/search:

`mcp` · `model context protocol`

Everyday Workflow

- Switch to plan mode with `Shift+Tab` before big changes.
- Review every diff and commit between tasks.
- Run `/branch` before a risky experiment.
- `/rewind` restores Claude's file edits but not changes from shell commands.

A Typical Loop

Command	What it does
`/init`	Once per project
`/plan add a CLI`	Agree on a plan
`/diff`	Review the edits
`/code-review --fix`	Catch bugs early
`/clear`	Start the next task

Learn More on realpython.com/search:

`claude code` · `git`

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